

Microsoft Agent Framework

Building blocks for the next generation
of enterprise AI agents.

Open Source

MIT Licensed

Python & .NET Preview

The Two-Framework Dilemma

Semantic Kernel

ROOTS

Production workloads & C#/.NET enterprise architecture.

STRENGTHS

Heavy-duty multi-agent orchestration, robust security hooks, rigid abstractions.

FRICTION

Steep learning curve, managing complex Kernel objects, import heavy.

VS

AutoGen

ROOTS

Microsoft Research (MSR), quick prototyping, and Python.

STRENGTHS

Simple interfaces, rapid experimentation, highly active community.

FRICTION

Challenging to scale into deterministic, compliant enterprise production environments.

The Synthesis: Best of Both Worlds

Enterprise Rigor (SK)

Developer Agility (AutoGEN)

Microsoft Agent Framework

A single, unified SDK merging enterprise rigor with developer agility.

1. Unified Abstractions

Drastically reduced lines of code and simplified imports. The Kernel is gone.

2. Enterprise Control

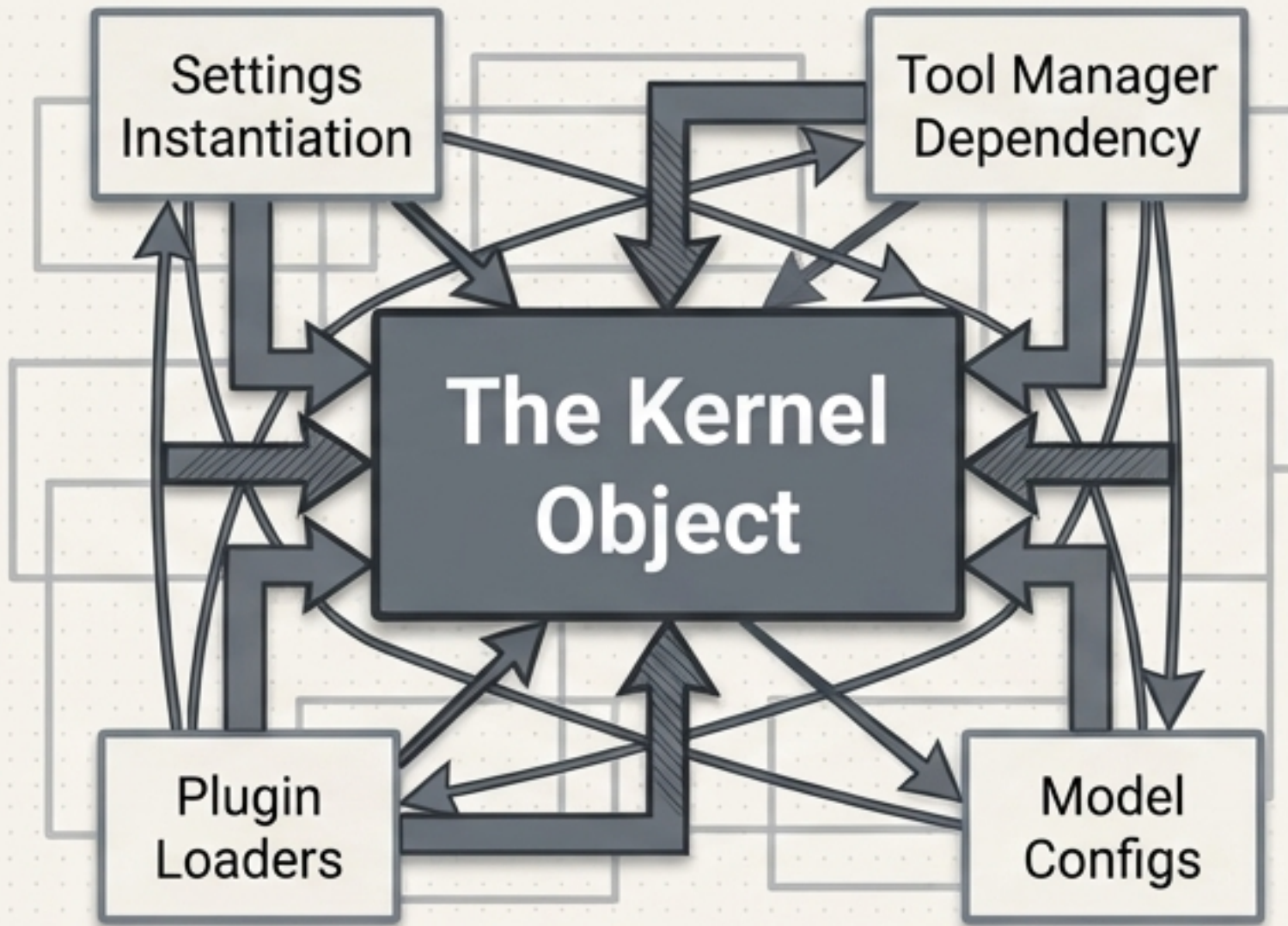
Tracing, guardrails, and deterministic workflows built-in for safe production.

3. Research Ready

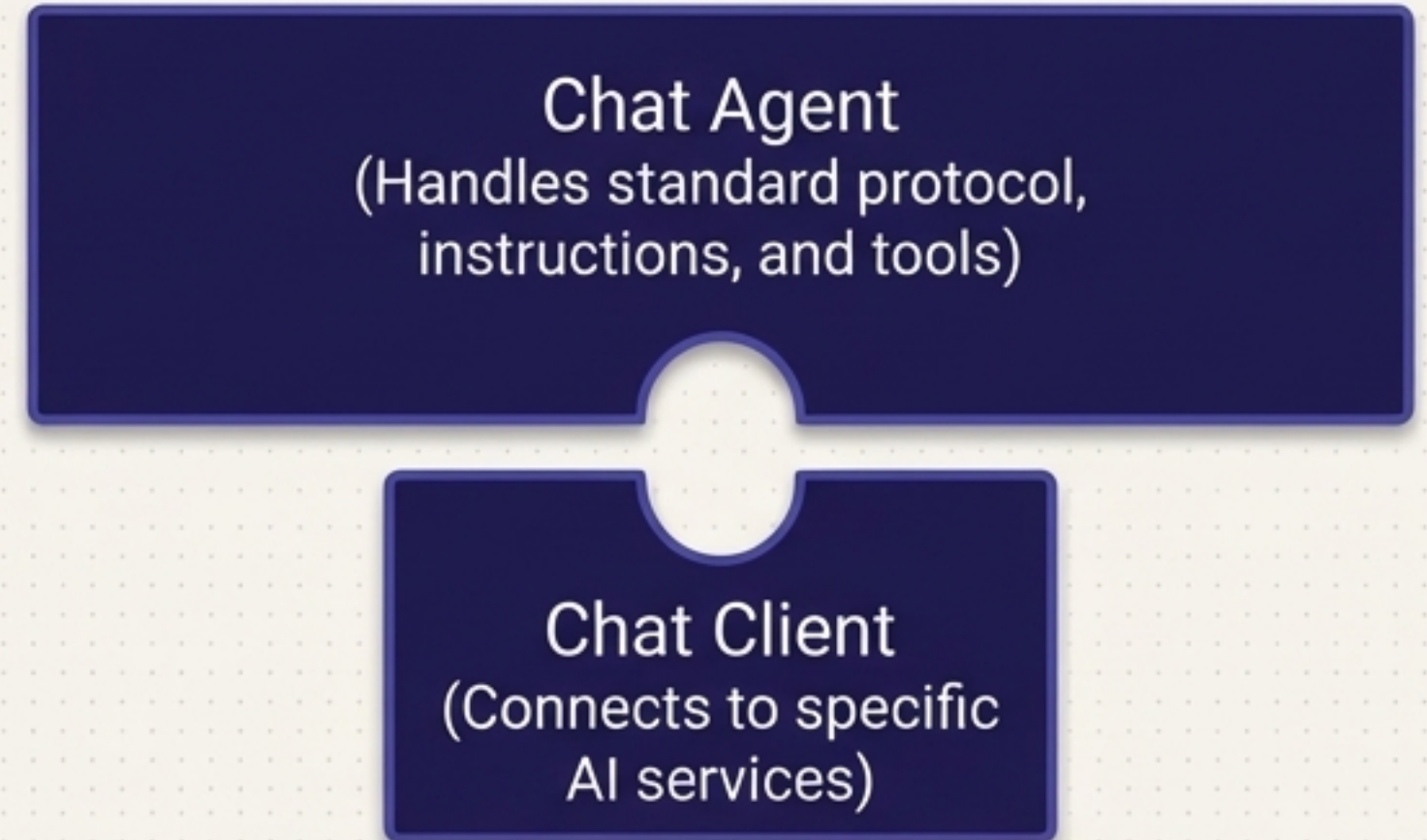
A dedicated lab package for experimental multi-agent patterns and MSR integration.

Unifying the Abstraction: The Kernel is Gone

Before (Legacy)



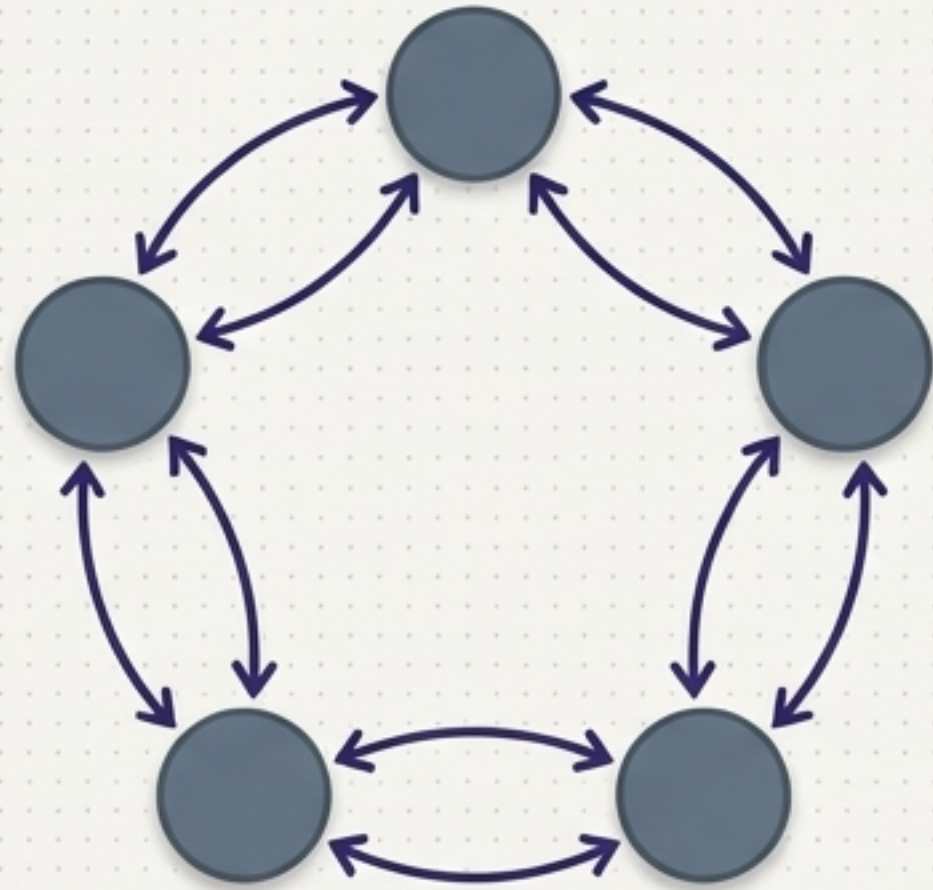
After (Agent Framework)



Stop worrying about which specific agent class to use.
Create a universal **Chat Agent**, drop in your **Chat Client**, and execute.

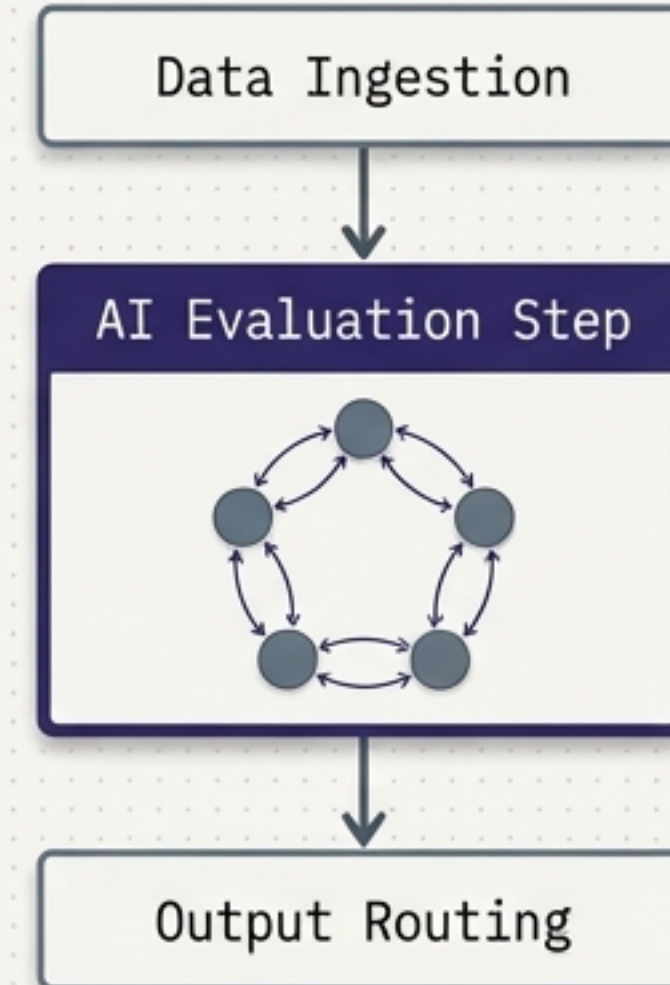
Orchestration: Autonomous vs. Deterministic

Agent-Driven Orchestration



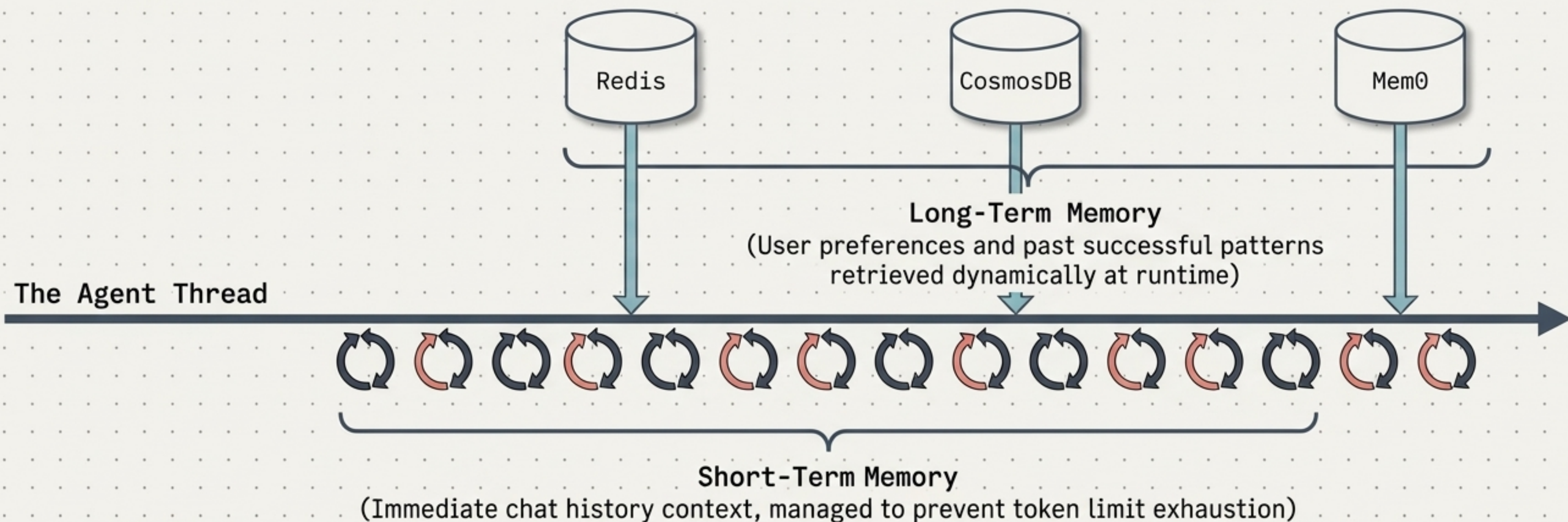
Supports familiar multi-agent communication models (e.g., the Magentic pattern, Concurrent pattern). Agents autonomously converse, pass state, state, and collaborate to complete complex tasks.

Workflow-Driven Orchestration



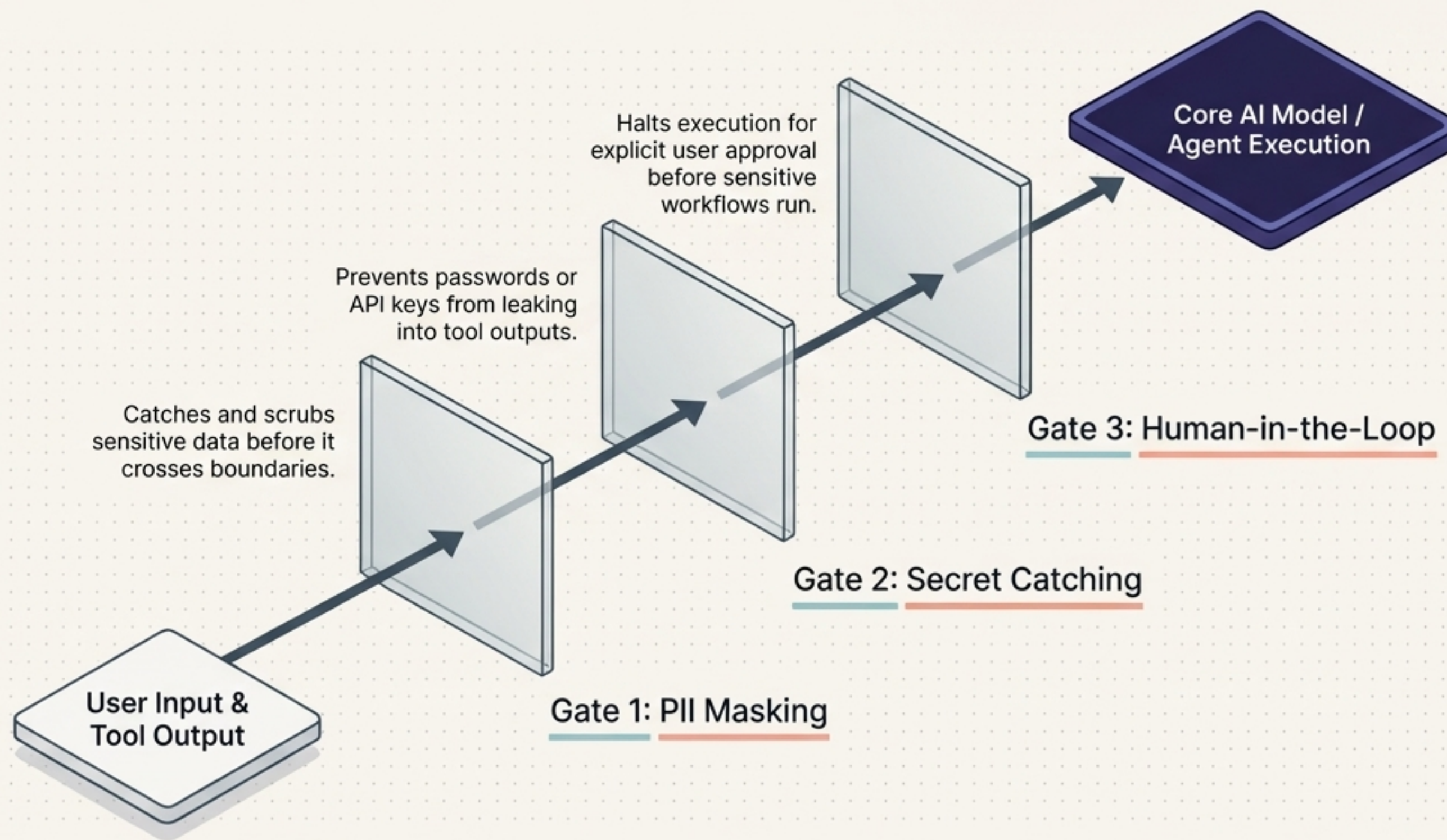
Define independent steps and transitions as a graph. Embed agents directly into traditional, deterministic business processes for enterprise reliability.

Rethinking Context: The Agent Thread



By separating the operational thread from external memory providers, agents retrieve context dynamically—preventing repeated mistakes and locking onto successful patterns.

Enterprise Control: Middleware & Guardrails



Previously known as "filters," middleware can now be cleanly injected at both the run-level and the agent-level, ensuring secure execution locally and in the cloud.

Developer Ergonomics: The Dev UI

Thread Management

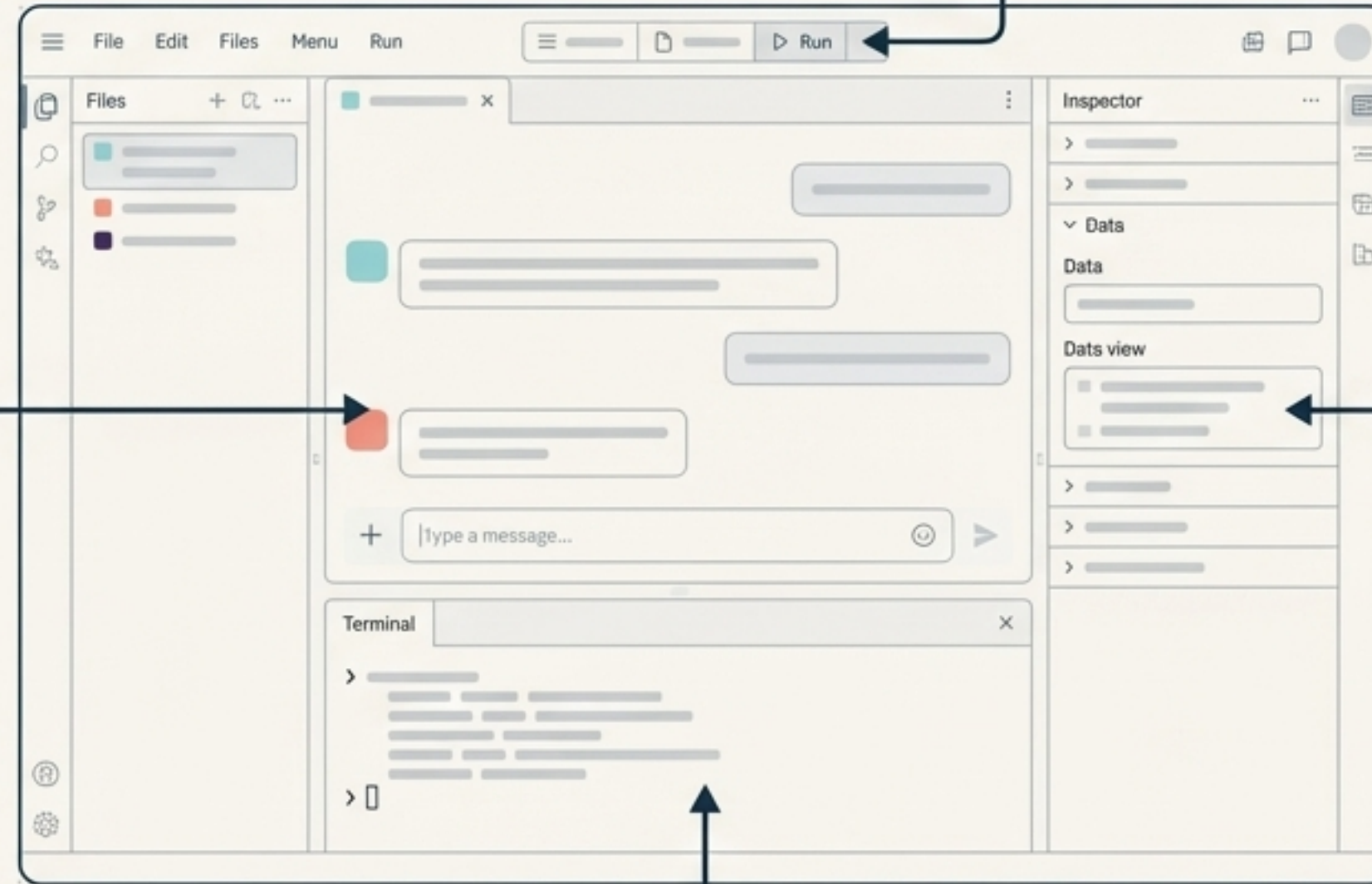
Visually switch between and manage multiple distinct agent conversation threads.

Multi-modal Support

Seamlessly handle vision/image inputs like screenshot transcription directly in the chat.

OpenTelemetry Integration

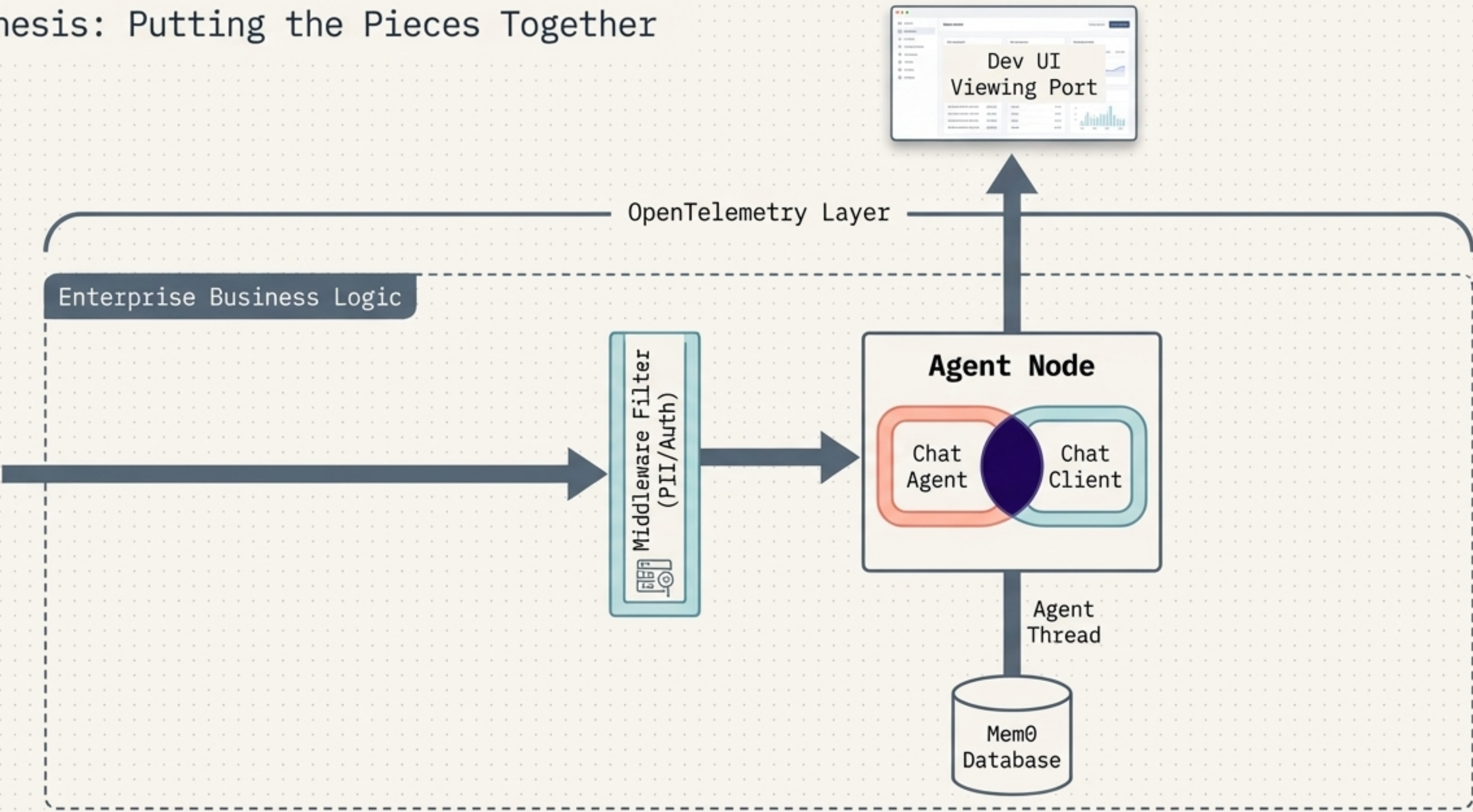
Interrogate detailed operational spans and tool calls cleanly in the side panel.



Local CLI Serve

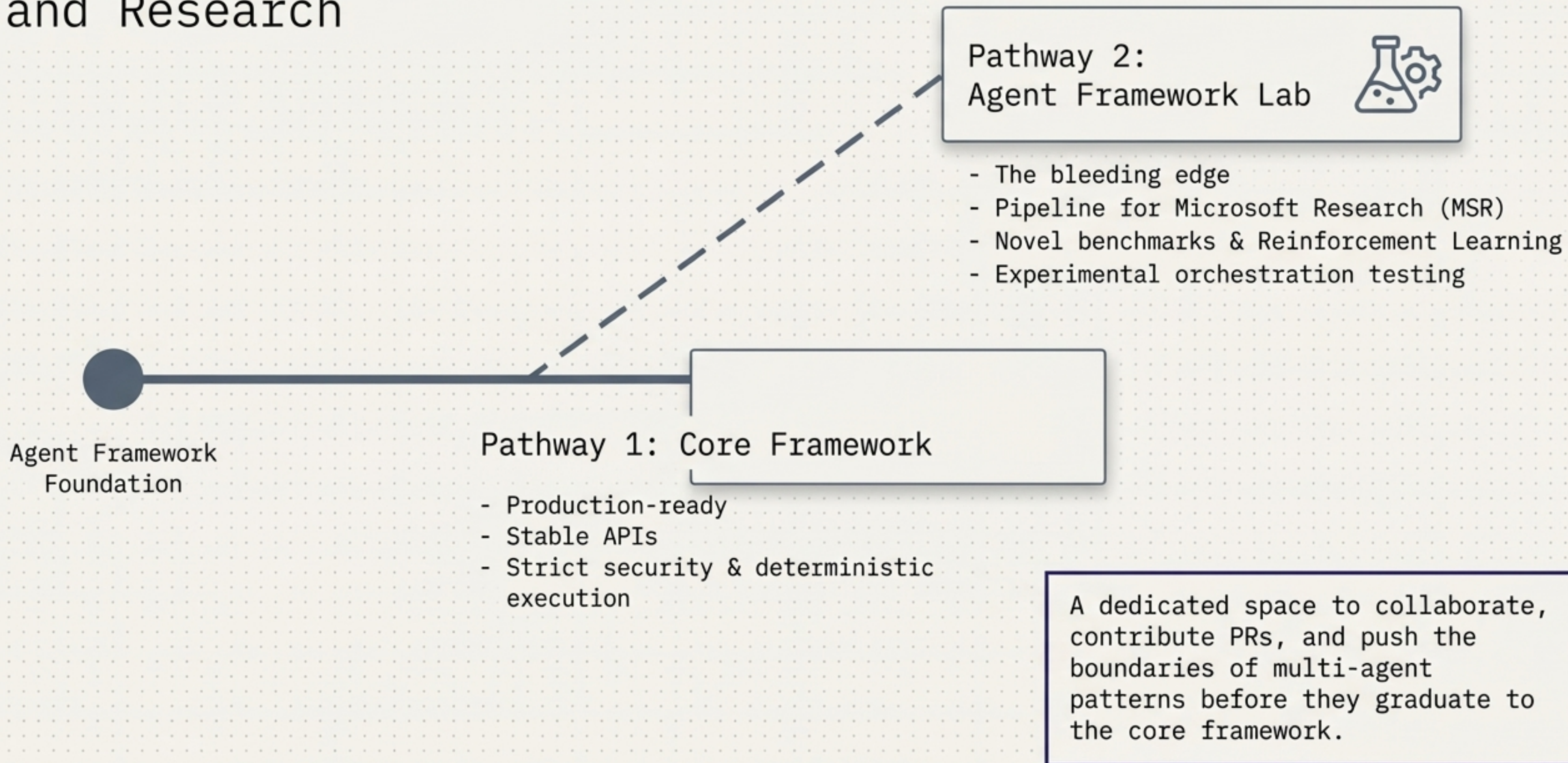
Point the CLI at a local folder to instantly serve all structured agents graphically.

Synthesis: Putting the Pieces Together



A fully observable, stateful, and secure AI layer, neatly encapsulated within deterministic business workflows.

Bridging Production and Research



The Developer's Stack (Preview)

Launch Languages

Python
(First-class support)

.NET
(First-class support)

Supported Integrations

OpenTelemetry

Mem0

Redis

CosmosDB

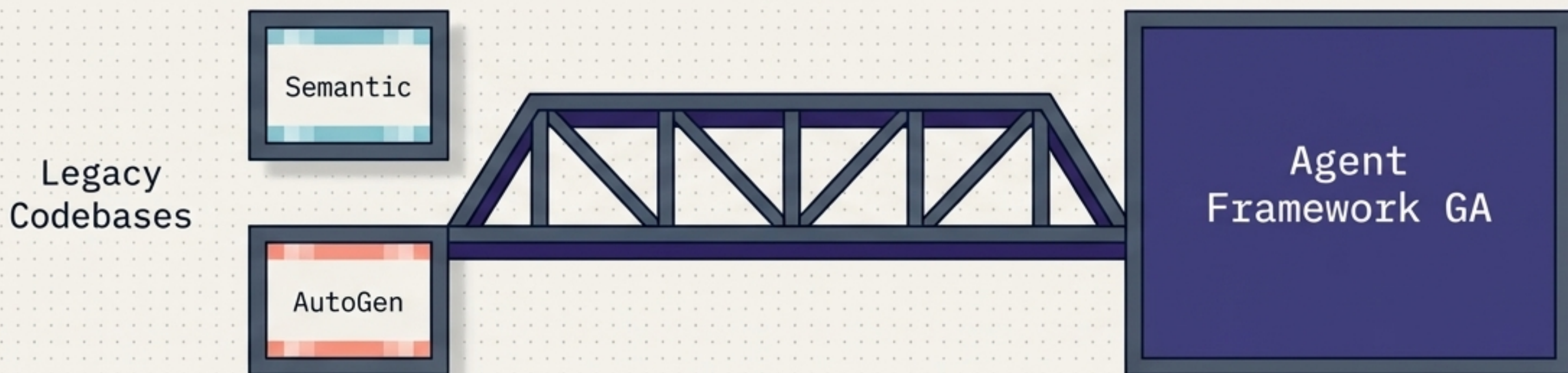
Roadmap Sneak Peek

Azure AI Foundry

Working closely with the Foundry team to provide fully managed agent memory services natively.

The Migration Promise & Next Steps

We know you've built extensively on AutoGen and Semantic Kernel. We are actively building comprehensive migration paths and documentation to ensure your transition is seamless. Your existing business logic will carry over.



Star the Repo ↗

Available now under MIT License on GitHub. Review the minimal-import Hello World implementation.

Open Issues ↗

Active community-driven development—your feedback shapes the GA release.

Join Office Hours ↗

Wednesdays at 8 AM PT. Meet the core team and bring your orchestration challenges.